

UMME FATIMA SADIA HOSSAIN

AI & BACKEND SOFTWARE ENGINEER

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Professional Summary

Software Engineering graduate building responsible AI and backend systems for healthcare, public services, risk intelligence and resilient communities. Experienced with Python, FastAPI, Django, RESTful APIs, PostgreSQL, RAG and multi-agent workflows, with a research interest in evidence grounding, uncertainty and human oversight.

Education

BSc (Hons) Software Engineering - Dual Award | Asia Pacific University & De Montfort University | **Completed Feb 2026**
Kuala Lumpur, Malaysia | Programme commenced Mar 2022 | English-medium instruction

Professional Experience

Project Management Intern | APU Career Centre | May 2024 - Aug 2024

- Completed a 16-week internship coordinating career-fair and student-programme operations across internal teams and external stakeholders, maintaining schedules, data accuracy and delivery milestones.
- Produced website prototypes and interface improvements for stakeholder engagement, while resolving operational bottlenecks through structured Agile collaboration.

Projects

GharDisha AI | USAII Global AI Hackathon 2026 Finalist | React | FastAPI | RAG

- Built a multilingual, source-grounded case interpreter that turns disaster-displaced families' housing stories and document/photo inputs into PMAY-G preparation plans while preserving human authority boundaries.

Sentinel Web-Risk Intelligence | Multi-agent third-party risk platform | FastAPI | CrewAI | Bright Data

- Designed a six-agent investigation workflow for live reconnaissance, verification, prediction and auditable reporting, supported by ChromaDB and containerized services.

Gamified Habit Tracker | Final-year research project - submitted 30 Apr 2025 | Django | PostgreSQL | Docker

- Engineered a research-informed habit system with authentication, configurable goals, streaks, badges, reminders, Chart.js insights and automated tests.

SenoNav AI | Evidence-aware healthcare navigation prototype | Python | Streamlit | RAG

- Developed a structured workflow for report ingestion, triage, referral readiness and multilingual, safety-first guidance with explicit human escalation.

Leadership and Service

Resident Assistant | The Purple House Hostel / INTI International College Subang | Oct 2019 - Nov 2020

- Supported international and first-year residents through transition programmes, community activities, incident reporting and day-to-day pastoral guidance.

Certifications and Awards

USAII Global AI Hackathon Finalist (Undergraduate, 2026) | Cisco AI Technical Practitioner | Cisco AI Business Practitioner
| AI Agent Olympics and Web Data Unlocked hackathon completions

Technical Skills

Backend: Python, FastAPI, Django, RESTful APIs, PostgreSQL, SQLite

AI and data: RAG, machine learning, multi-agent workflows, Gemini, Qwen, CrewAI, Google ADK, ChromaDB, GIS

Engineering: Docker, Git/GitHub, CI/CD pipelines, unit testing, Agile, Kanban, AWS, React, Next.js, Streamlit, JavaScript